# Paramdeep Singh Gill

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## Education

**Thapar Insitute of Engineering and Technology (TIET)** — Patiala, Punjab
*Bachelor of Engineering in Computer Engineering, CGPA: 8.09* — *Sep 2021 – Jun 2025*

**Apeejay School, Mahavir Marg (C.B.S.E. affliated)** — Jalandhar, Punjab
*10 + 2 (Non Medical), Percentage: 95.60%* — *2021*

**Apeejay School, Mahavir Marg (C.B.S.E. affliated)** — Jalandhar, Punjab
*10, Percentage: 90.33%* — *2019*

## Experience

**IIT Ropar** — Rupnagar, Punjab
*Research intern* — *January 2025 - June 2025*

- Explored Knowledge Distillation for Model Compression in Medical Domain.
- Under the guidance of Dr. Deepti Reddy Bathula (Indian Institute of Technology (IIT) Ropar, India)

**AgNext** — Sahibzada Ajit Singh (S.A.S.) nagar, Punjab
*Android developer intern* — *May 2023 - August 2023*

- Flutter: Built android apps that connect to BLE devices.

## Awards and Recognition

**Runner-Ups - HackForSustainability** — 2023
*Organised by Saturnalia Fest (largest in North India) for Project Saarthi* — *Patiala, Punjab*

**Winner - Smart India Hackathon (SIH) (Senior Software Edition)** — Aug 2022
*Team Leader and Front-End Developer (Problem Statement under DoSEL, Ministry of Education)* — *Karnataka*

**Awarded Merit Scholarship – II (Tuition fee waive off)** — 2022
*From Thapar Insitute of Engineering and Technology* — *Patiala, Punjab*

**Dr Stya Paul Award for Human Values – 2020** — 2020
*In Senior Secondary Category and scholarship at Apeejay School, Mahavir Marg* — *Jalandhar, Punjab*

**ATL Student Innovator Award** — 2019
*Certificate of Merit by Atal Innovation Mission, Niti Aayog* — *Jalandhar, Punjab*

**Qualified for National Championship of World Robot Olympiad (WRO)** — 2016 and 2017
*For the year 2016 & 2017. Secured third position in Regional Championship of WRO 2017.*

## On-going Projects

**XAI-FL** — 2026
*Research project* — *Remote*

- Trying to integrate XAI (Explainable Artificial Intelligence) on models trained using FL (Federated Learning). With a focus on using privacy preserving techniques like Differential Privacy.
- Guide: Dr. Parteek Kumar Bhatia (WSU, USA) and Dr. Sachin Kansal (TIET, India)
- Technology used: PyTorch, InterpretML, NVFlare.

**BioFL** — 2026
*Research project* — *Remote*

- Working on applying Federated Learning (FL) on unexplored biomedical datasets.
- Guide: Dr. Parteek Kumar Bhatia (WSU, USA) and Dr. Kashish Goyal (TIET, India)
- Technology used: PyTorch, InterpretML.

**Deepfake Detection** — 2026
*Group Research project* — *Remote*

- Working on building a model to detect deepfake in dataset having audio video modality.
- Guide: Dr. Shefali Arora Chouhan (NIT Jalandhar, India)
- Technology used: PyTorch

## Projects

**ThaparSat research project** 2024
*Solo project* *Patiala, Punjab*

- Systematic Approach to Enhance Signal Quality and Noise Reduction in Remote Sensing and Astrophysical Applications
- Capstone project under the guidance of Dr. Seema Bawa and Dr. Mamta Gulati

**Vriddhi App** 2022
*Group Project* *Patiala, Punjab*

- An app to guide parents to facilitate and ensure holistic growth in cognitive, psychomotor and affective domains. Contributed in the front-end of the app (using Flutter-Dart).
- Technology used: Flutter-Dart, Firebase, GetX, React Js, Chart Js, Swiper Js, Figma.

## Technical & Research Skills

**Machine Learning & AI**: PyTorch, InterpretML, NVFlare, OpenCV, NumPy, Matplotlib, Knowledge Distillation, Federated Learning
**Programming Languages**: C, C++, Python, Dart, HTML, CSS, SQL
**Research & Academic Tools**: LaTeX, AsciiDoc, Zotero, Git, GitHub, VS Code, JupyterLab
**Development Frameworks**: Django, Flutter, Firebase

## Extra-Curricular and Volunteer

**WRO 2026 Regional Championship, Jaipur** 8 August, 2026
*Judged the participants in 'Future Innovators' category.* *Jaipur, Rajasthan*

**WRO 2026 Regional Championship, Chandigarh** 8 July, 2026
*Judged the participants in 'Future Innovators' category.* *SAS nagar, Punjab*

**International Conference on Statistical Learning (ICSL)** 2025
*Presented 'Systematic Approach to Enhance Signal Quality...'* *Rupnagar, Punjab*

**Open Web Application Security Project (OWASP)** 2022 - 2023
*Team Member: Presented a session on blockchain as part of orientation of first year students.* *Patiala, Punjab*

**Center for Training and Development (CTD)** 2021 – 2023
*Senior Coordinator: Coordinated the Placement Symposium.* *Patiala, Punjab*

**International Award for Young People (IAYP)** 2019
*Successfully completed Bronze Category Course.* *Jalandhar, Punjab*

## References

**Dr. Deepti Reddy Bathula** bathula@iitrpr.ac.in
*Associate Professor, IIT Ropar* *Rupnagar, Punjab*

**Dr. Seema Bawa** seema@thapar.edu
*Professor, TIET* *Patiala, Punjab*